

		Important libraries:							
		NumPy → Numerical calculations, arrays							
		Pandas → Data cleaning, transformation, analysis							
		Matplotlib & Seaborn → Data visualization							
		Scikit-learn → Machine learning models							
		You don't need to build everything from scratch							
		Saves time in real-world projects							
		3. Efficient Handling of Large Data							
		Python can work with:							
		Large datasets							
		Structured and unstructured data							
		Apache Hadoop							
		Apache Spark							
		4. Strong Data Visualization Capabilities							
		Visualization helps in decision-making.							
		Using libraries like:		You can create:					
		Matplotlib		Bar charts					
		Seaborn		Line graphs					
				Heatmaps					

QUESTION 2		Explain the difference between List and Tuple in Python						
ANSWER		List						
		A list is a collection of items that is ordered and mutable (can be changed).						
		my_list = [1, 2, 3, 4]						
		Tuple						
		A tuple is a collection of items that is ordered but immutable (cannot be changed).						
		my_tuple = (1, 2, 3, 4)						
		Tuples are faster than lists						
		Lists are slightly slower due to flexibility						
		Use tuple when:						
		Data should NOT change						
		Performance matters						
		List = Flexible (changeable)						
		Tuple = Fixed (unchangeable)						

		4. Improves Readability					
		Code becomes clean and understandable					
		5. Easy Maintenance					
		If you need to change logic, update in one place					
		Automatically updates everywhere					
		6. Used in Data Analytics					
		Functions help:					
		Clean data					
		Perform calculations					
		Automate tasks					
		A function = reusable block of code					
		Helps in:					
		Saving time					
		Reducing errors					
		Making code clean and organized					

QUESTION 4		Write a Python program to take a user's name as input and print a greeting message.					
ANSWER							
		Take input from user					
		name = input("Enter your name: ")					
		Print greeting message					
		print("Hello", name + "! Welcome!")					
		input() → takes input from the user					
		name → stores the user's name					
		print() → displays the greeting message					
		Enter your name: Mitesh					
		Hello Mitesh! Welcome!					

QUESTION 5	Write a Python program to check whether a number is even or odd using conditional statements.
ANSWER	<pre># Take input from user num = int(input("Enter a number: ")) # Check even or odd if num % 2 == 0: print("The number is Even") else: print("The number is Odd")</pre> input() → takes number from user int() → converts input into integer % (modulus) → gives remainder If remainder is 0 → Even, else Odd Enter a number: 7 The number is Odd

QUESTION 6		Write a program to print numbers from 1 to 10 using a loop.				
ANSWER		for i in range(1, 11):				
		print(i)				
		range(1, 11) → generates numbers from 1 to 10			1	
		Loop runs for each number			2	
		print(i) → prints the number			3	
					4	
					5	
					6	
					7	
					8	
					9	
					10	

QUESTION 7	Create a list of five numbers and print the maximum number from the list.
ANSWER	Create a list of five numbers <pre>numbers = [10, 25, 7, 89, 45]</pre> Find and print the maximum number <pre>print("Maximum number is:", max(numbers))</pre> numbers → list containing 5 values max() → built-in function that returns the largest number output Maximum number is: 89

QUESTION 9		Write a function that returns the square of a number.			
ANSWER		def square(num):			
		return num * num			
		# Example usage			
		result = square(5)			
		print("Square is:", result)			
		def square(num):	→ defines the function		
		num * num	→ calculates the square		
		return	→ sends the result back		
		Square is: 25			

QUESTION 10	Write a Python program to count how many times a number appears in a list. Example List [2,3,4,2,5,2]
ANSWER	<pre>numbers = [2, 3, 4, 2, 5, 2] # Number to count num = 2 # Count occurrences count = numbers.count(num) print("Number appears", count, "times")</pre> numbers.count(num) → counts how many times the number appears in the list